SAMPLE

Famous Astronomers: George Gamow

The Big Bang Architect

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Introduction

In the middle decades of the twentieth century, no one made the universe more fun to think about than George Gamow. A Soviet-born physicist who became one of America's best-loved science writers, Gamow cracked the mystery of how atomic nuclei spit out particles, sketched the first serious picture of how the chemical elements were cooked in the fireball of a young universe, and predicted the faint glow of radiation that should still fill the sky if that fireball had really happened. Years before biologists worked out the details, he even took a serious run at the puzzle of how DNA's four chemical letters could spell out the recipes for every protein in a living cell.

George Gamow was born Georgiy Antonovich Gamov on March 4, 1904, in Odessa, a port city on the Black Sea in what was then the Russian Empire. His father taught Russian language and literature; his mother taught geography and history. The family was educated and multilingual, and young George grew up speaking Russian at home, picking up French from his mother and German from a tutor. He learned English only in his college years, but he mastered it so completely that it eventually carried both his technical papers and his celebrated books for general readers.

His scientific life stretched across some of the most turbulent and exciting decades in the history of physics. He studied in Leningrad under Alexander Friedmann, one of the first people to realize that Einstein's equations allowed an expanding universe. He worked in the great centers of European physics, in Gottingen, Copenhagen, and Cambridge, and then fled the Soviet Union in the tense early 1930s, landing in the United States, where he spent the rest of his career. Along the way he explained radioactive decay with the strange new rules of quantum mechanics, helped found the physics of the Big Bang, and charmed millions of readers with the adventures of a bank clerk named Mr Tompkins.

This book follows Gamow's life and work chapter by chapter. It begins with how the world discovered him, moves through the science that made him famous, examines how his boldest predictions were confirmed, and ends with the puzzles he left behind and the legacy that still bears his name. His story is one of curiosity, courage, and a stubborn insistence that the deepest questions in science deserve clear answers, and that those answers belong to everyone.

Chapter 1: The Discovery of George Gamow

Every scientific career has a moment when a private talent becomes a public fact, and for George Gamow that moment arrived in 1928, in Gottingen, when he solved a problem that had troubled physics for years: why do some radioactive atoms decay almost instantly while nearly identical atoms wait billions of years?

The puzzle was old and well documented. Early researchers studying radioactive materials had found that each kind of unstable atom has its own fixed half-life, the time needed for half of a sample to decay, and that the particles flying out of a nucleus during alpha decay carried certain characteristic energies. By the 1920s experimenters had even noticed a tidy pattern, later called the Geiger-Nuttall law, connecting the energy of the emitted alpha particle with how quickly the decay happened. High-energy emissions went with short half-lives; sluggish emissions went with staggeringly long ones. Nobody could explain why.

The classical picture made the whole thing impossible to understand. An alpha particle, a tight cluster of two protons and two neutrons, sits inside the nucleus held by an enormous nuclear force. To escape, it would need to climb over an energy barrier far taller than anything it could afford. By the old physics, an alpha particle lacking that energy should be trapped forever. Yet trapped particles escaped all the time, on a schedule as regular as a train timetable. Classical physics offered no mechanism whatsoever for how an imprisoned particle could just walk out of jail.

Gamow's insight was that the prisoners were not subject to the old physics at all. They lived by quantum rules, and quantum rules contain a loophole that no one had previously applied to the nucleus: there is always a small, calculable probability that a particle trapped behind a barrier can pass through it, as if the wall were a little like fog rather than solid stone. The new wave mechanics, barely two years old at the time, allowed what experimenters called tunnelling. Gamow worked out the theory of alpha decay as a tunnelling process during his time in Gottingen, with mathematical help from Nikolai Kochin, and he treated the model potential of the nucleus carefully enough to derive, from first principles, the observed relationship between half-life and emission energy. The empirical Geiger-Nuttall law, until then a naked fact, suddenly had a physical explanation. The same problem was solved independently, at almost the same moment, by Ronald Gurney and Edward Condon in Cambridge, though

Gamow's treatment reached the more precise quantitative agreement with the measured decay rates.

This was the discovery that made Gamow's name. The probability factor he uncovered, which describes a charged particle's chance of slipping through the electric repulsion barrier around a nucleus, became known in physics as the Gamow factor, and it would go on to matter far beyond radioactivity, because the same tunnelling trick is what allows stars to shine. He was twenty-four years old, fresh out of a doctorate built on studies of the atomic nucleus in Gottingen, and already he had touched one of the fundamental processes of nature.

Recognition came quickly. Gamow spent the years from 1928 to 1931 at the Theoretical Physics Institute of the University of Copenhagen, the intellectual home of Niels Bohr and the quantum revolution, with a working break alongside Ernest Rutherford at the Cavendish Laboratory in Cambridge. In Copenhagen he continued his nuclear work, proposing the liquid drop model of the nucleus, the first mathematical model of the nucleus as a whole, and he also turned his attention to the stars, working on stellar physics with Robert Atkinson and Fritz Houtermans. In 1931, at the age of twenty-eight, he was elected a corresponding member of the Academy of Sciences of the USSR, one of the youngest in its history.

That same year, 1931, brought a darker turn. Soviet authorities refused Gamow permission to attend a scientific conference in Italy. It was a small administrative refusal, but it told him which way the wind was blowing, and it set in motion the escape story that fills a later chapter of this book. The physicist who had just explained why particles can escape their prisons was beginning to plan an escape of his own.

The wider significance of the tunnelling discovery took only a few years to appear. By the close of the 1920s it was understood that the same quantum loophole that lets an alpha particle leave a uranium nucleus also lets two positively charged nuclei, which classically should repel each other forever, meet and fuse at the temperatures found inside stars. What began as an answer to a puzzle about radioactivity turned out to be the master key to nuclear reactions of every kind, in the laboratory and in the sky alike. Not bad for a doctoral student's insight, and the world of physics registered the arrival. His Gottingen work provided the basis for his doctorate, and the invitations that followed, Copenhagen, Cambridge, the collaborations with Atkinson and Houtermans on stellar physics, read like a tour of every institution that mattered in the new physics. By 1931, still in his twenties, he had been elected to his country's Academy of Sciences, and the escape years that followed only confirmed what the discovery had already established: here was a talent the world would have to reckon with.